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Sentiment Analysis of Online Shopping Platform Product Reviews

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Abstract—This project presents a sentiment analysis of Amazon product reviews using machine learning classifiers and optimization techniques. Two models were implemented: a Random Forest Classifier and a Decision Tree optimized using Grey Wolf Optimization (GWO). After preprocessing and applying TF-IDF for feature extraction, reviews were classified as positive or negative. The Random Forest showed strong baseline accuracy, while GWO significantly enhanced the Decision Tree's performance. The results, evaluated through classification reports and confusion matrices, highlight the value of combining traditional classifiers with metaheuristic optimization. The study encourages future research into multi-parameter tuning and comparisons with other swarm intelligence techniques.

Index Terms—Sentiment Analysis, Random Forest, Decision Tree, Grey Wolf Optimization

I. INTRODUCTION

This project focuses on analysing Amazon product reviews using sentiment analysis techniques. It applies Random Forest and Decision Tree classifiers, with the latter enhanced using Grey Wolf Optimization (GWO). By optimizing model performance, the study aims to improve classification accuracy and efficiency in understanding customer opinions from text data.

A. Background

Sentiment analysis helps e-commerce platforms interpret customer opinions from vast online reviews. This study enhances machine learning models—Random Forest and Decision Tree—using Grey Wolf Optimization for hyperparameter tuning, improving classification accuracy on large-scale Amazon product review datasets.

B. Problem Statement

With the surge in online reviews, manual sentiment analysis is impractical. This project employs Random Forests and Grey Wolf Optimization-enhanced Decision Trees to improve classification on large-scale Amazon review data.

C. Contributions

This paper contributes:

- Build a hybrid sentiment analysis model.
- Efficient hyperparameter optimization.

II. RELATED WORK

Recent advances can be categorized as follows:

A. Machine Learning Techniques

Sobia Wassana et al. [1] and Joy Chandra Gope et al. [2] discusses the examination of online social networks specially Amazon user reviews.

B. Hybrid rule-based approach

The paper by Anjali Dadhich et al. [3] deals with creating an automatic comment analyser and classification model for customer reviews.

C. Lexicon-based approach

Experimentation done by Vasundhara Raj et al. [4] and Zeenia Singla [5] have been done using methods like SentiWordNet, fuzzy logic and Naïve Bayes.

III. METHODOLOGY

Our simplified version comprises of four stages:

A. Data Collection and preprocessing

The dataset containing global product reviews is loaded and preprocessed by lowercasing, removing punctuation and stop words, and tokenizing text to reduce noise and help models focus on meaningful patterns.

B. TF-IDF Vectorization

Post preprocessing, text data is transformed into numerical format using TF-IDF vectorization, which highlights significant words in reviews by reducing the impact of commonly used terms and emphasizing relevance.

C. Model Training

Two models are built: a Random Forest for baseline performance and a GWO-optimized Decision Tree. Their classification accuracy is evaluated using metrics like precision, recall, F1-score, and confusion matrices.

D. Analysis and Deployment

Finally, results are analysed to assess GWO's benefits in hyperparameter tuning, showcasing a balanced integration of traditional machine learning and modern optimization techniques for enhanced sentiment classification on large-scale text data.

IV. RESULTS

- The Random Forest Classifier served as a strong baseline, offering high accuracy and consistent performance. Its ensemble strategy reduced variance and overfitting. Grey Wolf Optimization Refined Decision Tree accuracy to around 83%, especially excelling in Class 1 (positive) sentiment detection, showing strong precision, recall, and F1-score across iterations.

V. CONCLUSION

The sentiment analysis system demonstrated strong performance with Random Forest and GWO-optimized Decision Tree models. Future improvements may involve tuning additional hyperparameters, using diverse datasets, and exploring advanced models like BERT for better semantic understanding and accuracy.

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